STREPTOMYCIN

Ototoxic (vestibular)

Deep IM

CHEMICAL NAME O-2-deoxy-2-methylamino-α-L-glucopyranosyl-(1→2)-O-5-deoxy-3-C-formyl-α-L-lyxofuranosyl-(1→4)-N,N′-diamidino-D-streptamine (as sulfate)

CLASS Aminoglycoside antibiotic (antituberculosis)

TRADE NAMES Streptomycin sulfate injection · Strepto-Fatol

THERAPEUTIC USES / INDICATIONS

- Tuberculosis — second-line/resistant cases; formerly part of retreatment regimen.
- Brucellosis (with doxycycline), plague, tularaemia.
- Enterococcal endocarditis (with penicillin, if gentamicin resistant).

ADMINISTRATION

- Deep IM injection; rotate sites. (IV only in special cases.)
- Wear gloves when preparing (contact dermatitis in staff).

DOSAGES

- **Adult:** 15 mg/kg IM once daily (max 1 g); reduce dose in patients >60 yr.
- **Brucellosis:** 1 g IM daily for 2–3 weeks.
- **Children:** 20–40 mg/kg once daily (max 1 g).
- **Intermittent (2–3 times weekly):** 25–30 mg/kg (max 1.5 g); reduce in renal impairment.

PHARMACOKINETICS

- **Peak:** IM 1–2 h. Not absorbed orally.
- Not metabolised; crosses placenta.
- Excreted unchanged in urine. **t½:** 2–3 h.

PHARMACODYNAMICS

- Binds **30S ribosomal subunit** → misreading → bactericidal.
- Active against extracellular *M. tuberculosis*, *Brucella*, *Yersinia*, *Francisella*.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain and abscess.
- Perioral paraesthesia.
- Rash, fever.

ADVERSE EFFECTS (SERIOUS)

- Vestibular ototoxicity (vertigo, ataxia, nystagmus) > cochlear (hearing loss).
- Nephrotoxicity (less than other aminoglycosides).
- Neuromuscular blockade.
- Anaphylaxis; exfoliative dermatitis.

CONTRAINDICATIONS & CAUTIONS

- Pregnancy (fetal deafness).
- Myasthenia gravis.
- Aminoglycoside hypersensitivity.
- **Caution:** renal impairment, elderly, hearing loss.

KEY INTERACTIONS

- **Loop diuretics, other aminoglycosides, vancomycin:** ↑ oto/nephrotoxicity.
- **Neuromuscular blockers:** enhanced block.

NURSING MANAGEMENT

- Check pregnancy status.
- Baseline audiometry and vestibular assessment; monitor regularly.
- Monitor renal function.
- Ask about dizziness, unsteadiness, ringing in ears.
- Rotate IM sites; wear gloves.

PATIENT EDUCATION

- Report dizziness, unsteadiness, ringing in ears or hearing loss.
- Use effective contraception during treatment.
- Attend all injection appointments.

KEY POINTS

- Aminoglycoside used for TB and brucellosis.
- Contraindicated in pregnancy.
- Vestibular toxicity (vertigo).
- Deep IM; staff wear gloves.

SOURCES CHECKED **[218]** Streptomycin, IM (US label)

Full references p. 432–457